

Root Cause Analysis of an AI Safety Near-Miss: IBM Watson for Oncology (WFO)

What Happened?

In 2018, an investigation by STAT News revealed IBM Watson Health documents showing that Watson for Oncology (WFO), an AI-powered clinical decision support system, generated multiple examples of unsafe and incorrect cancer treatment recommendations. The internal presentation raised concerns about the system's treatment recommendations and the methods used to develop its clinical knowledge base. One physician from Jupiter Hospital in Florida reportedly told IBM that the software was unsuitable for use in many clinical cases because of the model's questionable recommendations. One example cited in the internal documents involved Watson recommending a 65-year old patient with newly diagnosed lung cancer and evidence of severe bleeding was recommended bevacizumab (Avastin), despite the drug carrying a box warning for severe or fatal hemorrhage in such patients. Although the investigation did not identify a confirmed patient injury directly caused by Watson's recommendations, it is documented as a significant AI safety near-miss, demonstrating the incorrect AI-generated recommendations could have resulted in patient harm if accepted without appropriate clinical oversight.

Why did it happen?

The primary issue was the way WFO was developed and validated. According to the internal IBM documents reported by STAT News, portions of the system's knowledge base were developed and tested using synthetic/hypothetical patient cases rather than real-world patient data. The document also indicates that the recommendations were based largely on the expertise of a small number of clinicians for each cancer type rather than extensive external validation against diverse clinical practice or real-world patient populations.

Evidence from Zou et al., 2020, a concordance study evaluating WFO in cervical cancer further illustrates these limitations. Overall treatment concordance between WFO and real clinical practice occurred in 72.8% of cervical cancer cases. However, disagreements frequently reflected differences in regional clinical practice. For example, Watson recommended treatments that were not medical reimbursement in China and did not include some therapies that were already

adopted in local oncology practice These findings demonstrate that an AI system developed within one healthcare environment may not generalise effectively to different healthcare systems.

What did the System Fail to Anticipate?

WFO failed to anticipate the complexity and variability of real-world clinical practice. It underestimated how differences in treatment guidelines, available medication, reimbursement policies and how patient characteristics would affect the suitability of its recommendations across different healthcare systems. The system also failed to account for the rapid change for oncology, where treatment guidelines and evidence continually change. Without continuous updating and independent external validation, AI-based clinical decision support systems risk becoming inconsistent with current standards of care.

Technical or Human-Factor Failure?

This incident is both technical and organisational failures. On a technical aspect, the system relied on limited training and validation methods that reduced its ability to generalise across diverse clinical settings. Organisationally, the system was promoted and developed before independent evidence demonstrated its performance across a wide range of real-world clinical environments. The case also highlights a human-factor issue: AI should support, not replace clinical judgement. Clinician oversight is essential to prevent automation bias and critically evaluate AI recommendations.

What would have caught it?

Several safeguard methods could have reduced the likelihood of this near-miss. Firstly, WFO should have undergone independent external validation using large, diverse real-world patient datasets before widespread implementation. Secondly, its clinical knowledge base should have been continuously updated to reflect and support the evolving clinical oncology guidelines and regional standards of care. Finally, the healthcare organisations should ensure that AI systems function strictly as decision support tools, with qualified clinicians reviewing all recommendations before treatment decisions. These safeguards would improve both the technical reliability of AI systems and their safe integration into clinical practice.

References

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